Frequency Range 3 for ISAC in 6G: Potentials and Challenges

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***Abstract*—Spanning 7–24 GHz, frequency range 3 (FR3), is a key enabler for next-generation wireless networks by bridging the coverage of sub-6 GHz and the capacity of millimeter-wave bands. Its unique propagation characteristics, such as extended near-field regions and spatially nonstationary fading, enable new transmission strategies. This article explores the potential of FR3 for integrated sensing and communication (ISAC), which unifies wireless communication and environmental sensing. We show that FR3's bandwidth and multiple-input multiple-output (MIMO) capabilities enable high-resolution sensing, multi-target tracking, and fast data transmission. We emphasize the importance of ultra-massive MIMO with extremely large aperture arrays (ELAAs) and the need for unified near-field and far-field channel models to support efficient ISAC. Finally, we outline challenges and future research directions for ELAA-based ISAC in 6G FR3.**

I. Introduction

Sixth-generation (6G) wireless technologies are expected to integrate communication and sensing capabilities within a unified framework. Integrated sensing will not only enhance network performance but also enable new applications such as autonomous driving, indoor localization, and environmental monitoring and imaging. A key design goal of *integrated sensing and communication* (ISAC) [1]–[3] is the development of unified waveform and beamforming strategies that jointly serve both sensing and communication, improving spectral, energy, and cost efficiency. These integration gains, however, critically depend on the optimal design of such waveforms and beamformers, tailored to the distinct propagation characteristics of the communication and radar channels within the frequency bands envisioned for 6G.

The *upper mid-band frequency range*, spanning 7.125 GHz to 24.25 GHz, also referred to as *frequency range 3 (FR3)* or centimeter-waves (cmWaves), has been identified as a new band for 6G networks [4]–[8]. This range bridges the gap between the previously defined FR1 and FR2 bands used in earlier generations of cellular networks, as illustrated in Fig. 1 and defined below.

- **FR1:** 410 MHz–7.125 GHz (used in 2G to 5G; includes sub-6 GHz bands)
- **FR2:** 24.25 GHz–71 GHz (used in 5G millimeter-wave)
- **FR3:** 7.125 GHz–24.25 GHz (fills the gap between FR1 and FR2; proposed for 6G)

But why should a new frequency band be introduced for 6G? What are the implications, potentials, and challenges of such a choice in ISAC? We address these questions throughout this article. To put things into perspective, we first discuss the historical frequency band assignments used in previous generations of cellular networks.

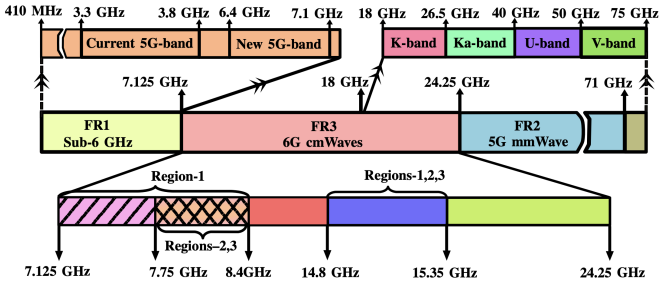


Fig. 1: FR3 and existing cellular frequency bands (FR1 and FR2) and their respective spectrum allocations.

For decades, the FR1 (which includes *sub-6 GHz band*) has dominated mobile communications, including key frequencies used by 3G (850, 1900, and 2100 MHz), 4G (850, 1700/2100, 1900, and 2300 MHz), and 5G (3.3–3.8 GHz). The FR1 band offers several advantages, including wide-area coverage, strong signal penetration through buildings, and low hardware and deployment costs. However, FR1 suffers from severe spectrum congestion and lacks wide contiguous bandwidths, limiting its ability to support ultra-high data rates.

5G was designed to markedly enhance capacity, peak data rates, connectivity, and latency compared to 4G. In particular, it aimed to deliver up to $1000\times$ higher data volumes, 10–$100\times$ more device connections, 10–$100\times$ faster user data rates, and $5\times$ lower latency than 4G operating in FR1. To achieve its ambitious data rates, 5G integrated advanced multi-user multiple-input multiple-output (MIMO) techniques with access to wide contiguous bandwidths within the *millimeter-wave (mmWave)* spectrum, more precisely in FR2 (24.25 GHz–71 GHz).

Despite attracting significant research attention over the past decade, FR2 suffers from several major limitations, including reduced coverage, high penetration losses through buildings and foliage, and increased complexity and cost compared to FR1. A primary concern is that, mmWave signals struggle to penetrate obstacles such as walls, windows, and vegetation, unlike FR1 signals. Moreover, the extremely narrow beams required by mmWave antenna arrays pose challenges for maintaining reliable connectivity under mobility. As such, mmWave has seen limited success as a mobile connectivity solution both in the U.S. and worldwide. However, fixed wireless access based on mmWave has proven to be effective as an alternative to wired broadband in several use cases [9]. The global uptake of mmWave remains modest, as countries like South Korea and most of Europe have largely deprioritized mmWave due to deployment complexity and limited mobile use cases.

To overcome the above challenges and balance coverage

and data rates, FR3 has been identified as a key candidate for 6G [4]–[8]. Referred to as the "golden band," FR3 has wider bandwidths than FR1, and offers more favorable propagation than FR2 due to lower path/penetration losses. Nokia's recent coverage study [10] showed that the lower FR3 (7 GHz) can achieve comparable or better cell-edge throughput than 5G at 3.5 GHz (FR1) when equipped with larger antenna arrays. The upper FR3 (13 GHz) also supports reasonable coverage using directional antennas and more transmit chains at the user equipment (UE) and customer premises equipment (CPE) [10].

As the bandwidth allocated to 6G in FR3 is limited—unlike the abundant bandwidth available in FR2—a natural solution is to exploit ultra-massive MIMO by deploying a very large number of antennas. This leads to the concept of *extremely large aperture arrays (ELAAs)*. ELAA-assisted wireless communication is thus expected to become a cornerstone for implementing ISAC in FR3 band. We argue that, due to the large physical aperture and high antenna count of ELAAs, the conventional *far-field* channel assumptions used in MIMO systems (such as those in the 5G FR1) no longer hold. Hence, it is imperative to investigate *near-field* communication models and techniques. The distinct propagation characteristics in the near field will introduce new challenges and demand the development of novel transmission and reception techniques.

This paper explores the potential of FR3 in enabling efficient ISAC for 6G. Our contribution can be summarized as

- **FR3 spectrum for ISAC:** We discuss available spectrum and challenges, including co-channel interference with existing satellite services, and highlight the need for agile spectrum access and management techniques.
- **Ultra-MIMO with ELAAs for ISAC:** We examine the need for ultra-MIMO with ELAAs to realize ISAC gains in FR3 and demonstrate how its unique propagation can be leveraged to provide seamless coverage along with high spectral and energy efficiency.
- **ISAC implementation in FR3:** We emphasize the need for unified far-field and near-field channel models that account for angle and distance dependencies. We advocate learning-based solutions to address the real-time design challenges of ultra-MIMO ISAC in FR3.

II. The golden band (FR3) and its existing services

Following the World Radiocommunication Conference 2023 organized by the ITU, specific FR3 sub-bands were identified as 6G candidate frequencies [5]. In Region 1 (Europe, Africa, Mongolia, and the Middle East west of the Persian Gulf), the candidate band is 7.75–8.4 GHz; in Regions 2 and 3 (the Americas, most of Asia, and Oceania), it is 7.125–8.4 GHz. Additionally, the 14.8–15.35 GHz band was identified for use across all three regions. In Region 2, U.S. service providers could access up to 1825 MHz of spectrum for 6G deployment.

A. Incumbent Communication and Sensing Services in FR3

Spectrum regulatory agencies worldwide have identified numerous existing civilian and government services operating in FR3. These incumbent services include fixed wireless systems, space research, radio astronomy, aeronautical and maritime radio navigation, non-terrestrial networks such as meteorological and Earth exploration satellites, maritime mobile and broadcast satellites, unmanned aerial systems, satellite feeder links, and broadband satellite Internet services.

According to the Federal Communications Commission, the largest spectrum allocations in the lower portion of FR3 (7.125–8.4 GHz) are held by fixed wireless and fixed/mobile satellite services for federal use [4]. For example, the Federal Aviation Administration operates microwave links in this band to reliably connect air traffic control centers with remote aeronautical radionavigation radars. Additionally, defense satellite communications systems use geostationary satellites operating in this range, with 7.25–7.7 GHz allocated for downlink and 7.9–8.4 GHz for uplink of defense and wideband gapfiller satellites. Non-federal uses of the lower FR3 band include unlicensed allocations for ultra-wideband devices operating in the 7.75–8.75 GHz range. These devices serve several consumer sectors, including personalized item tracking, wall-penetrating radars, automotive radars, and wearable technology.

The upper portion of FR3 (8.5–13.75 GHz) hosts several incumbents, with the largest allocation for radiolocation services [4]. Bands within 12.2-13.75 GHz may be available for spectrum sharing with full and limited access. Non-federal uses include the downlink and uplink of commercial fixed/mobile satellite services. Earth stations and user gateways use these bands for broadband Internet, serving enterprises, government entities, emergency responders, schools, and hospitals. For example, Starlink primarily operates in Ku (12–18 GHz) and Ka (27–40 GHz) bands, using 10.7–12.7 GHz for downlink and 14 GHz for uplink. Federal uses include NOAA's geostationary and data relay satellites, NASA's active sensors, precipitation missions, and radars, and NSF-supported radio astronomy. Internationally, FR3 supports X-band (8–9 GHz) radars for civil and defense applications, land/naval/military radars, and fixed/mobile satellite services.

B. Coexistence of 6G ISAC Functionalities with Incumbents

A key challenge in adopting FR3 for ISAC in 6G lies in addressing regulatory considerations, particularly enabling spectrum sharing and managing co-channel interference through advanced antenna arrays and signal processing. As mentioned, many government and commercial communication, satellite, radar, and navigation services operate in both the lower and upper portions of FR3 for mission-critical applications.

Terrestrial ISAC systems can coexist with these satellite services without significant performance degradation by employing advanced spatial interference cancellation techniques. To enable the coexistence, ISAC base stations (BSs) must employ unified waveforms and beamformers that jointly optimize communication, sensing, and spectrum sharing by maximizing sum rate and sensing accuracy under dynamic constraints, such as spatial interference nulling and permissible interference levels at satellite UEs and CPEs to ensure compatibility within FR3.

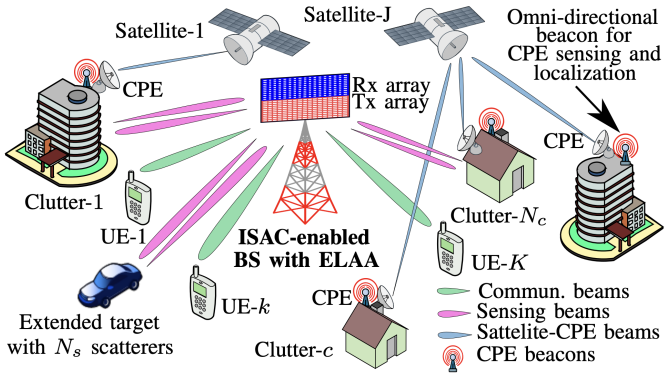


Fig. 2: Coexistence of UEs and satellite CPEs in FR3 with a BS using an ELAA.

C. Challenges of Unification and Coexistence of ISAC in FR3

The ISAC unification is constrained by the coexistence with a myriad of incumbents, particularly including satellite services in FR3 and underlying optimization procedures are also challenging. This is because design objectives of the communications, sensing, and spectrum sharing tasks are mutually coupled and competing each other. To meet data rates envisioned for 6G, innovation of ultra MIMO technologies in FR3 will also be needed [5]. As such, model-based multi-objective optimization and signal processing techniques may not be useful for designing 6G ISAC with incumbents in FR3 as the optimal solutions may not be feasible due to model inadequacy or optimal solutions may be too complicated for real-time implementation. To this end, data-driven artificial intelligence-empowered end-to-end design approaches may hence be appealing.

The above challenges are compounded by the need for accurate sensing of satellite CPEs operating in the FR3 band to enable ISAC coexistence (see Fig. 2). Co-channel interference must be efficiently managed, which requires spatial interference nulling toward satellite CPEs while preserving ISAC functionality. This, in turn, demands precise CPE localization, dynamic spectrum sensing, and channel estimation. While classical pilot-based techniques can estimate BS-UE channels, passively sensing satellite CPEs is extremely difficult due to their highly directional links.

Passive sensing of satellite CPEs appears practically unrealizable, even for modern BSs, but active sensing is feasible through pilot-based UE-BS channel estimation. A pragmatic solution involves beacon/tone-based active sensing, where satellite CPEs periodically transmit omni-directional beacons. With mutual agreements between 3GPP and satellite providers, ISAC-enabled BSs can use these beacons for accurate localization, angle-of-arrival estimation, and dynamic spectrum sensing. Efficient sensing of satellite CPEs in FR3 for ISAC coexistence remains an open research challenge.

III. Candidate Technologies for 6G ISAC in FR3

This section elaborates candidate technologies to overcome the key challenges of designing and implementing ISAC with incumbent services in FR3.

A. ELAAs for Enabling ISAC in FR3

Co-located massive MIMO technology is currently serving as one of the key enablers of 5G deployments. The 5G BSs operating in lower FR1 band (3.5 GHz) are typically equipped with 16, 32 or 64 antenna arrays. They adopt *time-division duplexing* with full digital processing such that each antenna is connected to a separate RF chain. Since mmWave 5G is not suitable for extensive coverage, it is typically used as a supplementary service in dense urban areas with high traffic. Aggressive spatial multiplexing is not typically needed for mmWave thanks to large bandwidth availability in this band, and high data rates can still be attained through efficient beam-searching and beam-steering techniques. Fixed wireless access is also a major use-case of 5G mmWaves. The apertures of arrays used for 5G mmWaves are much smaller than those used for the 5G sub-6 GHz band for the same number of antenna terminals.

There are two proven techniques to linearly increase the data rates, i.e.,

1) using wider bandwidths in unoccupied spectrum, and
2) leveraging spatial multiplexing gains rendered by large antenna arrays.

The first technique is currently being utilized by 5G mmWave BSs to boost data rates as there is an ample amount of usable bandwidths within FR2. This is because the achievable rate (R) is proportional to the communication bandwidth (W) due to Shannon's channel capacity theorem: $R = W \log_2(1+\gamma)$, where γ is the signal-to-noise ratio (SNR). However, the propagation in FR2 is not favorable to provide seamless coverage and connectivity for cellular services, and this band is being used mostly for fixed wireless access. Most 5G cellular systems around the world have been deployed in FR1, and total bandwidth allocation for 5G in this sub-6 GHz band is about 1200 MHz (see Fig. 1).

Interestingly, the total bandwidth availability for 6G in FR3 is comparable to that of 5G in FR1. While the maximum allocation in FR3 reaches 1825 MHz across all regions, Region 1 (including the U.S.) is allocated only 1200 MHz. Thus, relying solely on wider bandwidths may be insufficient to meet 6G data rate targets, given the similar overall bandwidth availability.

The second technique to linearly increase the data rates involves transmitting multiple data streams to serve many users in the same time-frequency resource element by using spatial multiplexing with antenna arrays. This technique is particularly appealing to increase the spectral efficiency (η), defined as the data rate per unit bandwidth. Specifically, η is a linearly increasing function of the spatial multiplexing rate, $\min(M, K)$, where M is the number of antennas at the BS and K is the sum of antennas at all UEs. Increasing the number of antennas, specially at the BS, is thus a practical way to boost data rates for a given bandwidth. Packing an enormous number of antennas gives rise to ELAAs consisting of densely-packed electrically-small antennas relative to the wavelength [11]. We henceforth categorize ELAA-aided wireless communication to the broad category of ultra MIMO, which will be the foundation of implementing ISAC in 6G within FR3.

B. Challenges of Implementing ISAC with ELAAs in FR3

Current MIMO systems for communication and radar services are based on two exclusive assumptions:

i. all UEs and radar targets are located in the *far-field* of the antenna array, and
ii. the channels exhibit wide-sense stationary (WSS) properties in the spatial domain.

Thus, existing designs assume that channel models with locally planar wavefronts over the antenna array are valid, and that the entire channel is visible to all UEs/targets. The accuracy of these classical far-field channel models has been thoroughly verified for the FR1 band through both channel measurement campaigns [12] and theoretical limits [13].

For all practical transmission distances in the FR1 band, it is guaranteed that almost all UEs/targets will be located in the far field of the antenna array. This is because the *Fraunhofer array distance* (d_{FA}), which demarcates the boundary between the radiative *near-field* and *far-field* regions of an antenna array, is on the order of a few meters. This follows from the expression $d_{\mathrm{FA}} = 2Nd_a^2/\lambda$ [11], where d_a is the largest dimension of an antenna element, N is the number of antennas in a planar array, and λ is the operating wavelength.

The adoption of far-field channel assumptions for MIMO system designs is therefore well justified for 5G in the FR1 band, as the underlying antenna arrays are electrically small (i.e., their dimensions are small relative to λ). However, the same cannot be stated for 6G systems operating in the FR3 band, due to fundamental changes in antenna array design and propagation characteristics.

In Fig. 3a, the variation of d_{FA} with the aperture size of the ELAA (D) and the number of antenna elements N is shown for two carrier frequencies in the FR3 band. The ELAA is modeled as a uniform square planar array consisting of N tiny antenna elements, each with horizontal and vertical dimensions of $\lambda/2$. The spacing between adjacent elements is also $\lambda/2$, making D directly proportional to $\sqrt{N}$. Figure 3a reveals that d_{FA}, and hence the near-field region, can extend to hundreds of meters when ELAAs are deployed in the FR3 band.

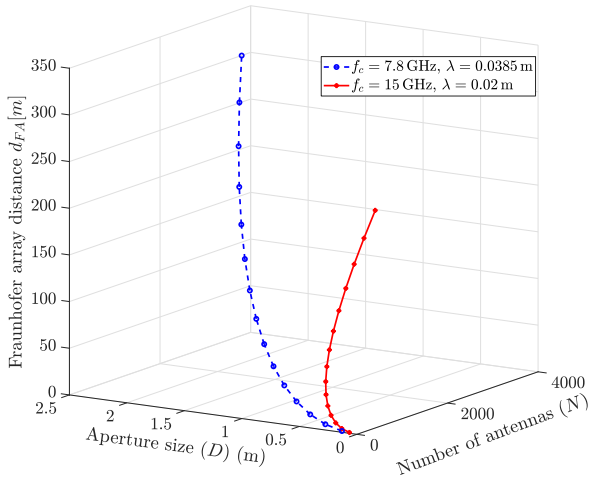
Under the far-field or planar-wavefront assumption commonly used in prior work, the array response—including steering vectors in communications and target response matrices in radar—depends only on the angles of arrival and departure (AoA/AoD). The far-field *radar cross section* (RCS) depends solely on the extended target size and wavelength [14]. The key implications of spatially WSS assumption are that the statistical properties of the received signals/echoes, such as the average received power and AoDs/AoAs, do not vary over the array's spatial dimensions. Via this WSS assumption, the following simplifications have been invoked into the existing ISAC models and designs; (i) the statistical mean of the spatial channel coefficient is a constant across the array, and (ii) the covariance depends only on the difference between the antenna indices. Another implication is that the existing ISAC designs rely on the fact that UEs and targets are visible to the entire array and thus paving the notion of wholly-visible scatterers and full visibility regions in the existing ISAC channel models [12].

6G BSs in FR3 must use large co-located or distributed ELAAs to achieve the high spectral and energy efficiency needed, as FR3 offers similar bandwidth to FR1 but suffers higher path loss. Previously, we observed that the Fraunhofer array distance is precisely N times larger than the Fraunhofer distance of an individual antenna. Hence, the combination of ELAAs and higher frequency bands undermines the fundamental assumptions that the wavefronts of radiated electromagnetic waves are locally planar over the antenna arrays and the underlying channels are spatially WSS. Hence, the adoption of ELAAs in FR3 largely extends the radiative near-field of the antenna array due to increased aperture sizes and operating frequencies (see Fig. 3a). Consequently, the UEs and sensing targets may be located predominately within the radiative near-field. Such wireless systems exhibit fundamental changes in electromagnetic properties and propagation characteristics. Wireless transmission and reception with ELAAs must also be explored by using integrals over the continuous apertures instead of the conventional approach of summations of individual antennas.

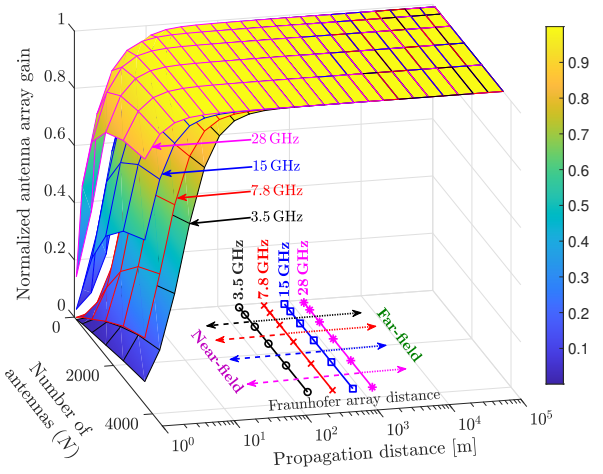
One of the primary purposes of ELAAs is to adopt beamforming to achieve a much larger total channel gain than that with a single antenna element. With far-field assumption, the effective array gain typically increases linearly with the number of antennas, and it is independent of the propagation distance (d_p). Hence, the antenna and array gains can be decoupled in the far-field. However, the gains of an antenna and the array in the near-field must be treated jointly, and it exclusively depends on d_p. Figure 3b shows the variations of the array gain of an ELAA against d_p and N in the array. Specifically, the normalized array gain, which is computed as a ratio between the total received power of the ELAA with N antennas and the received power of N references antennas, is plotted while keeping the arrays aperture size (D) the same for the frequencies 3.5 GHz in FR1, 7.8 and 15 GHz in FR2, and 28 GHz in FR3. Since D is fixed, the Fraunhofer array distance increases with carrier frequency. It is seen that the normalized array gain varies with d_p and N. This clearly justifies the need to capture unique near-field propagation characteristics for efficient implementation of ISAC in FR3 band.

For ELAAs in FR3 band, the UEs and radar targets for automotive applications are likely to be located in the extended near-field. Rigorous measurement campaigns [12] have revealed that the statistical signal properties vary significantly over ELAAs, and spatial WSS assumption may no longer be valid. Consequently, the far-field planar wavefront and spatially WSS assumptions may no longer be valid for ELAAs. Thus, the existing physical-layer ISAC designed based on far-field channel models will become inaccurate/incomplete once ELAAs are adopted in FR3 band. The first key challenge is to establish theoretical foundations of unique ELAA channel features such as multiple depth-of-foci, non-WSS and visibility regions in spatial domain to implement ISAC in FR3 band. The second challenge is to discover efficient ISAC system designs to leverage unique ELAA near-field channel features to practically realize integration gains of ISAC designs.

Super-directivity and favorable propagation characteristics of ELAAs in FR3 will also lead to finer spatial resolutions



(a) Fraunhofer array distance against ELAA's aperture size and number of antennas.



(b) The normalized array gain against the propagation distance and number of antennas.

Fig. 3: ELAAs performance within FR3 band.

and significantly lower transmit power requirements. The combination of unprecedented spatial multiplexing gains and low power requirements will pave the way to realize (i) large spectral/energy efficiency gains for communication tasks and (ii) boost estimation, detection and localization accuracy for sensing tasks in ISAC systems subject to fundamental design trade-offs for 6G operating in FR3.

The extremely large apertures and propagation characteristics of the FR3 band demand novel precoding/combining and signal processing techniques. Key challenges include interference rejection, frequency-selective beamforming to address beam squint, and dynamic multi-user precoding to handle spatial non-stationarity and varying visibility regions across the array. All aforementioned distinctions with respect to massive MIMO in FR1 band justify that radically novel channel/signal models, signal processing techniques and new performance analysis frameworks are necessary to explore the true potential of the ELAAs in FR3 band as an enabling ISAC technology in 6G.

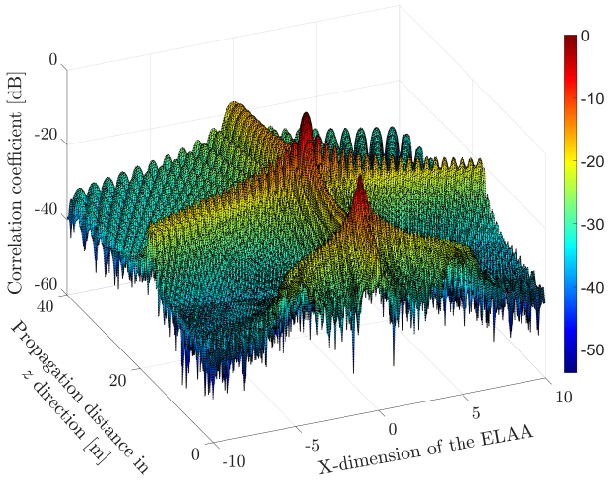


Fig. 4: Correlation coefficient in dB scale when one UE is fixed and a second UE is moved along the same spatial direction.

C. Implications of Extended Near-Field for ISAC in FR3

To efficiently implement ISAC in FR3 band, unified and generalized near/far-field channel models need to be discovered. Unique ELAA propagation features must also be leveraged to practically realize integration gains of ISAC. To this end, the ELAA can be modeled as a 2D transmit antenna surface with a densely-packed large number of antennas each having a finite area. These antennas are assumed to be placed edge-to-edge to constitute an extremely large contiguous antenna surface with an approximately continuous electrically-large aperture. Thus, the propagation effects must be investigated using 2D integrals over the entire aperture instead of typical summations of individual antennas. The channels between ELAA and UEs/targets must also be modeled via a superposition of spherical wavefronts by virtue of circular and divergent phase gradients to capture distinct propagation distances, effective antenna areas, and polarization mismatches in spatial/angular domains.

New near-field ISAC channel models must capture spatial finite depth-of-foci, non-WSS, small-scale, and large-scale fading effects. The array response vectors of both communication and radar channels depend not only on AoAs/AoDs in the elevation and azimuth planes but also on the transmission distances. Figure 4 shows that the array steering vectors of two UEs located in the same azimuth and elevation directions correlates and decorrelates repeatedly along the distance domain in the near-field of ELAAs. Hence, the beams transmitted by an ELAA possess multiple focal points with finite depths along the same spatial direction in the near-field as opposed to the infinite depth-of-focus of beams in far-field channel

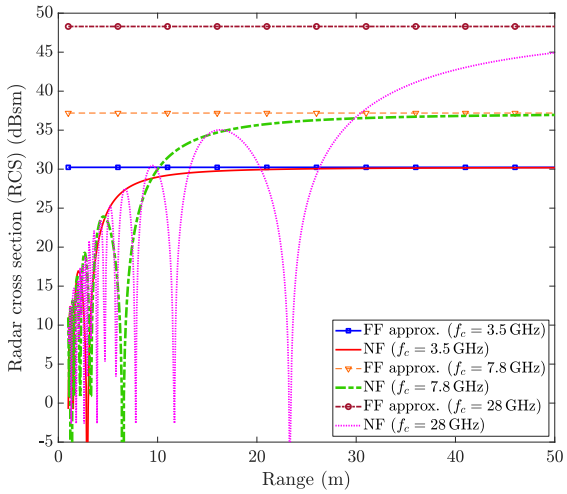


Fig. 5: The RCS versus range for different frequency bands with far-field (FF) and near-field (NF) assumptions.

models with plane-wavefronts. Multiple UEs/targets having the same spatial direction can hence be served/illuminated by using near-field beam-focusing. This task requires designing of unified beamformers and waveforms that can be jointly used for both communication and sensing functionalities within the same bandwidth allocated in FR3 band.

The target response matrix of a radar channel is a function of RCS, and the array steering vectors [14]. Classical definition of the RCS of an extended target exclusively relies on the far-field planar wavefront assumptions. Hence, the RCS becomes independent on the target range (r). For example, for a circular 2D target with the radius a, the RCS with far-field definition is given by $\sigma = 4\pi^3 a^4/\lambda^2$ [14]. However, Fig. 5 uncovers that once the electric field in the near-field ELAA channels is considered, the RCS will also depend on the range. Hence, the radar target response matrices in the near-field will depend on the range in addition to the AoAs/AoDs in the elevation and azimuth planes. In the far-field, the range and AoAs are estimated based on the time-delay and target response matrix [1], respectively. In the state-of-the-art sensing systems, the resolution of time-delay and time difference of arrival estimations are typically hindered by the bandwidth and synchronization requirements [1]. However, with accurate near-field channels for ELAAs in FR3 band, both the range and AoA of a target can be estimated directly through the range-angle dependent target response matrix without relying on the traditional time-delay-based estimators.

Accurate sensing of unique ELAA channel characteristics in the extended near-field of FR3 is essential for enabling ISAC integration gains through unified waveform and beamformer design–i.e., communication waveforms operating also in the delay-Doppler domain or combined communication/sensing waveforms. These characteristics can be captured via spatial-time, spatial-frequency, angle-delay, and delay-Doppler representations using 2D Fourier transforms. Delay-Doppler domain modeling is especially useful, as its parameters map directly to scatterers and allow embedding communication symbols in unified waveforms. ELAA's finite focal depths and high angular resolution enable novel multiplexing, allowing multiple UEs with similar angles but different ranges to be served simultaneously without interference. Additionally, ELAA super-directivity helps mitigate spatial aliasing, enhance radar resolution, and reduce clutter in 6G ISAC systems.

Near-field beamforming aims to tackle the interplay between near-field propagation and spatial-wideband effects, such as beam squint and split, in wideband ELAAs [11]. True-time delays (TTDs) can be incorporated into hybrid beamformer architectures to mitigate the joint impact of spatial-wideband fading effects in the near-field. The candidate beamformer design approaches are:

- *Sub-array partitioning*: A large array is divided into several smaller sub-arrays, each approximated as operating in the far-field planar region.
- *Application integration*: Near-field beamforming is combined with other emerging technologies such as reconfigurable intelligent surfaces (RIS) and holographic metasurface-based antenna arrays. For instance, RIS Fresnel zone properties can be exploited, with optimization adapted to TTD and analog beamforming constraints.
- *TTD configuration*: Exploring different TTD configurations, including parallel, serial, and hybrid architectures.
- *Deep learning-based methods*: This includes transformer-based and end-to-end learning-based methods.

IV. A USE-CASE OF FR3 FOR DEPLOYING ISAC IN 6G

We investigate a use-case of implementing ISAC in FR3 band with an ELAA-equipped BS. The antennas at the ELAA are arranged as a square UPA of size 1.243×1.243 m. Two carrier frequencies in FR3 band (7.8 and 15 GHz) are used, and the FR1 frequency 3.5 GHz is used as a baseline comparison. The size of an antenna element is $\lambda/4 \times \lambda/4$. The relative antenna spacing is set to $\lambda/2$. For the same form factor, the numbers of antennas in the ELAA for 3.5, 7.8 and 15 GHz are 400, 961, and 1521, respectively. The UEs, target, and clutter lie in the near-field, with distances below the Fraunhofer limit ($d_{FA} = 72.1$ m for $f_c = 3.5$ GHz). For 7.8 and 15 GHz, the limits are 160.65 m and 308.96 m, respectively. The BS-UE distances are set to $[30, 40, 50, 60]$ m. The extended target has four scatterers, and their distances are set to $[30, 30, 31, 31]$ m. The coherence interval τ_c is set to 196 [13], and the pilot length for uplink channel estimation is set to $\tau_P = K$. The uplink pilot transmit SNR is set to 0 dB.

In Fig. 6, the trade-off between the achievable rate per user ($\mathcal{R}$) and the probability of detection (P_D) is depicted for the optimal ISAC waveform designed for the near-field channel models for three carrier frequencies $f_c \in \{3.5, 7.8, 15\}$ GHz. The target detection is performed by adopting the detector proposed in [15, Eq. (69)], with a fixed false alarm probability of 10^{-7}. Six rate-detection trade-off curves are generated by varying the numbers of UEs and antennas at the ELAA as $K \in \{4, 6\}$ and and $N \in \{400, 961, 1521\}$, respectively. These curves are obtained by adjusting a weighting factor, ρ, which governs the priority assigned to communication versus

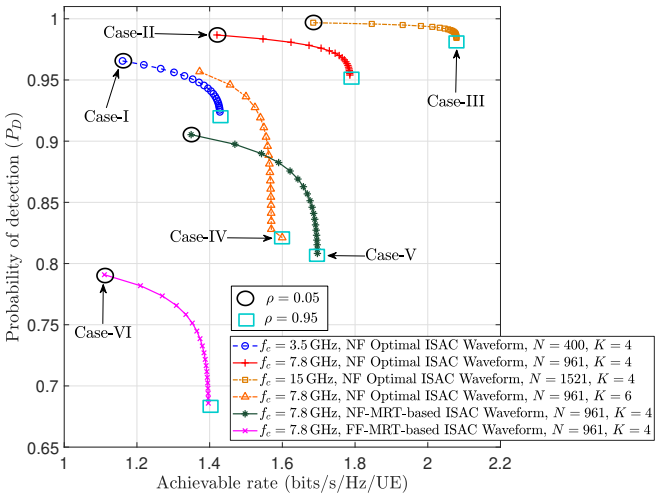


Fig. 6: The trade-off between the achievable rate and probability of detection.

sensing objectives during ISAC waveform design, as proposed in [2]. Specifically, ρ is varied from 0.05 to 0.95 in increments of 0.05 at the average transmit SNR of 10 dB. Each point along the trade-off curves corresponds to a distinct ρ value, indicating different ISAC operating points. Figure 6 illustrates the fundamental ISAC trade-off: as the communication rate $\mathcal{R}$ increases (higher ρ), the detection probability P_D decreases, indicating a shift toward communication at the expense of sensing.

In Cases I to IV, an optimal ISAC waveform is used, by jointly minimizing multi-user interference and the mismatch with a reference sensing waveform having good correlation properties [2]. Since the ELAA's form factor is kept fixed for all three frequencies, the number of $\lambda/4 \times \lambda/4$ antennas that can be packed in the UPA increases with higher frequencies. Hence, it is evident that increasing the number of antennas in the ELAA enhances both the UE rates and detection probability. An MRT-based ISAC precoder using an accurate near-field model (Case V) is compared against a far-field-based design (Case VI) to assess performance loss. Comparing Cases V and VI shows noticeable sum rate and detection losses when a far-field planar wavefront-based ISAC precoder is used when UEs are actually located in the near-field at 7.8 GHz (FR2). Furthermore, comparing Cases II and IV reveals degraded detection performance as the number of UEs increases.

V. Conclusion

We have investigated the potential of the FR3 for ISAC in 6G. Our work suggests that this "golden band" offers a unique opportunity to achieve wide-area coverage, high data rates, and high-resolution sensing. Toward this, we highlighted the importance of unified near- and far-field channel models, dynamic spectrum sensing and advanced transceiver designs for enabling seamless coverage and enhanced performance for ISAC in FR3. To fully exploit the propagation characteristics of FR3 band for ISAC, the use of ultra-massive MIMO with ELAAs a practically viable path forward.

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